

STROBE Statement — Checklist of items to be included in reports of cross-sectional studies

Manuscript: Olfactory dysfunction, daytime activity reduction, and 24-hour rhythm fragmentation — but not food-odor recognition — in NHANES 2013–2014

Author: Ceyhun Olcan

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This checklist follows the STROBE Statement for the reporting of cross-sectional studies (von Elm et al., *PLoS Med* 2007;4(10):e296). Page numbers refer to the v25 manuscript PDF.

Title and abstract

#	STROBE item	Recommendation	Reported on page	Section
1a	Title/abstract	Indicate the study's design with a commonly used term in the title or the abstract	1	Title; Abstract Methods
1b	Title/abstract	Provide in the abstract an informative and balanced summary of what was done and what was found	1	Abstract (Background, Methods, Findings, Interpretation)

Introduction

#	STROBE item	Recommendation	Reported on page	Section
2	Background/rationale	Explain the scientific background and rationale for the investigation being reported	1–2	Introduction (paragraphs 1–2)
3	Objectives	State specific	2	Introduction

#	STROBE item	Recommendation	Reported on page	Section
		objectives, including any prespecified hypotheses		(paragraph 3); Methods pre-specifications

Methods

#	STROBE item	Recommendation	Reported on page	Section
4	Study design	Present key elements of study design early in the paper	2	Methods — Population and exposure
5	Setting	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	2	Methods — Population and exposure (NHANES 2013–2014)
6	Participants	(a) Give the eligibility criteria, and the sources and methods of selection of participants	2; Figure 1	Methods — Population and exposure; STROBE flowchart (Figure 1)
7	Variables	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	2–3	Methods — Outcomes; Methods — Regression framework (M4 covariates)
8	Data	For each	2–3	Methods —

#	STROBE item	Recommendation	Reported on page	Section
	sources/measurement	variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group		Population (PST), Methods — Outcomes (MIMS, accelerometry), Tables 1–4
9	Bias	Describe any efforts to address potential sources of bias	3	Methods — Regression framework (IPSW, frailty adjustment, sensitivity analyses); Discussion — Limitations
10	Study size	Explain how the study size was arrived at	2; Figure 1	Methods — Population and exposure; STROBE flowchart
11	Quantitative variables	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	2–3	Methods — Outcomes; Methods — Regression framework
12a	Statistical methods	Describe all statistical methods, including those used to control for confounding	3	Methods — Regression framework
12b	Statistical	Describe any	3	Methods —

#	STROBE item	Recommendation	Reported on page	Section
	methods	methods used to examine subgroups and interactions		Regression framework (20 OD × modifier interactions); Results — Effect modification
12c	Statistical methods	Explain how missing data were addressed	3	Methods — Regression framework (MICE, complete-case primary)
12d	Statistical methods	If applicable, describe analytical methods taking account of sampling strategy	3	Methods — Regression framework (MEC weights, stratum × PSU clusters)
12e	Statistical methods	Describe any sensitivity analyses	3	Methods — Regression framework; Results — Sensitivity analyses

Results

#	STROBE item	Recommendation	Reported on page	Section
13a	Participants	Report numbers of individuals at each stage of study	Figure 1	Sample selection flowchart
13b	Participants	Give reasons for non-participation at each stage	Figure 1; 4	STROBE flowchart; Results — Sample
13c	Participants	Consider use of a flow diagram	Figure 1	STROBE flowchart
14a	Descriptive data	Give characteristics of study	Table 1	Sample characteristics by OD status

#	STROBE item	Recommendation	Reported on page	Section
		participants and information on exposures and potential confounders		
14b	Descriptive data	Indicate number of participants with missing data for each variable of interest	3; Methods	Methods — Regression framework (MICE detail); Results — Sample
15	Outcome data	Report numbers of outcome events or summary measures	Tables 2–3; Figure 3	Results — OD-associated activity signature
16a	Main results	Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (e.g., 95% confidence intervals). Make clear which confounders were adjusted for and why they were included	Tables 2–3; Figure 3	Results — Regression (M1 unadjusted through M4 fully adjusted)
16b	Main results	Report category boundaries when continuous variables were categorized	2	Methods — Population (OD ≤ 5 correct cutoff); Methods — Outcomes (Karas thresholds)
16c	Main results	If relevant, consider translating estimates of	4	Results — Comparison with self-report magnitude

#	STROBE item	Recommendation	Reported on page	Section
		relative risk into absolute risk for a meaningful time period		(BMI-category anchor)
17	Other analyses	Report other analyses done — eg analyses of subgroups and interactions, and sensitivity analyses	4–6	Results — Comparison with self-report magnitude through Sensitivity analyses (8 subsections)

Discussion

#	STROBE item	Recommendation	Reported on page	Section
18	Key results	Summarise key results with reference to study objectives	6	Discussion (paragraphs 1–2)
19	Limitations	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias	7	Discussion — Strengths and limitations
20	Interpretation	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and	6–7	Discussion (cholinergic hypothesis, comparison with Schubert et al.)

#	STROBE item	Recommendation	Reported on page	Section
		other relevant evidence		
21	Generalisability	Discuss the generalisability (external validity) of the study results	7	Discussion — Strengths and limitations (race/ethnicity stratification); Implication and forward

Other information

#	STROBE item	Recommendation	Reported on page	Section
22	Funding	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	8	Conflict of Interest and Authorship (funding declaration)

Additional notes

- **Pre-registration:** The follow-up mortality analysis (paper #2) was prospectively pre-registered at <https://doi.org/10.17605/OSF.IO/ZX8RN> before downloading the NHANES Linked Mortality File. The current cross-sectional manuscript was not formally pre-registered prior to data acquisition (acknowledged in Methods — Population and exposure).
- **Data sharing:** All raw data are publicly available from CDC NHANES at <https://wwwn.cdc.gov/nchs/nhanes/>. Analytic code, dataset reconstruction pipeline, and minute-level fragmentation extraction script are deposited at <https://github.com/ceyhunolcan/od-activity-rhythm> (Zenodo concept DOI 10.5281/zenodo.20132927).
- **Reporting guideline followed:** STROBE statement for cross-sectional studies (von Elm et al., *PLoS Med* 2007;4(10):e296).

This checklist was prepared per STROBE Statement v4 guidelines. The corresponding author affirms that all 22 STROBE items have been addressed in the v25 manuscript and supplementary materials.